GraphMamba: An Efficient Graph Structure Learning Vision Mamba for Hyperspectral Image Classification

Aitao Yang, Min Li, Yao Ding, Leyuan Fang, *Senior Member, IEEE*, Yaoming Cai, *Member, IEEE*, and Yujie He

***Abstract*— Efficient extraction of spectral sequences and geospatial information is crucial in hyperspectral image (HSI) classification. Recurrent neural networks (RNNs) and Transformers excel in capturing long-range spectral features, while convolutional neural networks (CNNs) excel in aggregating spatial information through convolutional kernels. However, RNNs and Transformers suffer from low-computational efficiency, and CNNs have limitations in perceiving global contextual information. To address these issues, this article proposes GraphMamba—an efficient graph structure learning vision Mamba for HSI classification. Specifically, GraphMamba is a novel hyperspectral information processing paradigm that preserves spatial–spectral features by constructing spatial–spectral cubes and employs a linear spectral encoder to enhance the operability of subsequent tasks. The core components of GraphMamba include the HyperMamba module, which enhances computational efficiency, and the SpatialGCN module, designed for adaptive spatial context awareness. The HyperMamba mitigates clutter interference by employing a global mask (GM) and introduces a parallel training and inference architecture to alleviate computational bottlenecks. Meanwhile, the SpatialGCN utilizes weighted multihop aggregation (WMA) for spatial encoding, emphasizing highly correlated spatial structural features. This approach enables flexible aggregation of contextual information while minimizing spatial noise interference. Notably, the encoding modules of the proposed GraphMamba architecture are both flexible and scalable, providing a novel approach for the joint mining of spatial–spectral information in hyperspectral images. Extensive experiments were conducted on three different scales of real HSI datasets. When compared with state-of-the-art classification methods, GraphMamba demonstrated superior performance. The core code will be released at https://github.com/ahappyyang/GraphMamba.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image (HSI) classification, mamba, remote sensing, state space model (SSM).**

Received 11 May 2024; revised 10 September 2024; accepted 23 October 2024. Date of publication 7 November 2024; date of current version 15 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62006240. *(Corresponding authors: Min Li; Yao Ding.)*

Aitao Yang, Min Li, Yao Ding, and Yujie He are with Xi'an Research Institute of High Technology, Xi'an 710025, China (e-mail: 824360083@qq.com; proflimin@163.com; dingyao.88@outlook.com; ksy5201314@163.com).

Leyuan Fang is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with the Peng Cheng Laboratory, Shenzhen 518000, China (e-mail: fangleyuan@gmail.com).

Yaoming Cai is with the School of Information Engineering, Zhongnan University of Economics and Law, Wuhan 430073, China, and also with the Emergency Management Research Center, Zhongnan University of Economics and Law, Wuhan 430073, China (e-mail: caiyaom@zuel.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3493101

I. Introduction

HYPERSPECTRAL images (HSIs) consist of many narrow and contiguous spectral bands. Compared with traditional panchromatic and multispectral remote sensing images, they have a broad spectral response range and high spectral resolution. Furthermore, hyperspectral images demonstrate impressive capabilities in identifying target coverage. They can detect materials with distinctive spectral absorption features, allowing for precise differentiation of land cover types, road pavement materials, and more. Countries worldwide attach great importance to the development of hyperspectral remote sensing, with its applications becoming increasingly widespread and achieving remarkable results in fields such as environmental monitoring [1], land cover classification [2], military target recognition [3], and precision agriculture [4].

HSIs contain rich information about land covers. However, the abundance of spectral bands leads to information redundancy and high interband correlation, which may result in the Hughes phenomenon. Additionally, HSIs often suffer from low spatial resolution, leading to mixed pixel effects. Relying solely on spectral information for feature extraction may lead to misclassification. These challenges make efficient processing of large datasets and accurate classification increasingly demanding.

To address the deficiencies of spectral information representation in supervised classification, researchers have proposed many methods for HSI classification over the past decade. These methods can be broadly categorized into two types: traditional methods and deep learning methods. Traditional methods focus on discriminative feature extraction and compact representation of spectral data, with representative methods including random forests [5], K-nearest neighbors (KNN) [6], and support vector machines (SVMs) [7]. However, these methods often treat the pixels in HSIs as independent spectral curves for classification, neglecting the spatial structural information, leading to low classification accuracy. Additionally, they lack robustness in the presence of noise and information loss.

With the rapid development of deep learning, its powerful nonlinear expressive capability has attracted wide attention in recent years. Compared with traditional methods, deep learning methods have achieved automatic extraction of high-level

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semantic information, thus avoiding cumbersome feature engineering. Recurrent neural network (RNN) is suitable for processing sequential data and can effectively capture discriminative features in sequences. Since the spectra can be regarded as sequential data, RNN can fully utilize HSI's high spectral resolution to capture long-range feature dependencies. For example, Zhou et al. [8] used cascaded RNNs to eliminate redundant information between spectral bands and simultaneously explored complementary information in spectral sequences. Zhang et al. [9] employed the local spatial sequential (LSS) method to extract low-level structural information, which was then passed to RNN for generating high-level semantic information. Compared with RNN, Transformer [10] can better capture global dependencies in sequences and possess high flexibility. Mei et al. [11] introduced the group embedding module in Transformer to achieve local-global spectral context feature extraction. To address the insufficient spatial feature exploration capability of Transformer, Yang et al. [12] proposed a joint GCN and Transformer network to capture subtle spectral differences. To jointly extract spectral, spatial, and semantic information from HSIs, Xie et al. [13] proposed a new semantic and spatial–spectral feature fusion Transformer (S3FFT) network, which achieves the integration of shallow spatial–spectral features and high-level semantic information. To address the issue of weak representations caused by the random generation of positional encoding in Transformers, Jiang et al. [14] proposed a new graph generative structure-aware Transformer (GraphGST), which is designed with a new absolute positional encoding to obtain the absolute positional sequences of pixels.

Moreover, convolutional neural networks (CNNs) [15], [16] have been widely applied in HSI classification due to their excellent spatial perception ability. To tackle the challenges of spectral information redundancy and insufficient spatial resolution, Chang et al. [17] designed a comprehensive CNN, where 2-D CNN was used to extract spatial semantic information, and 3-D CNN was responsible for extracting spectral discriminative features to reduce information redundancy. Ye et al. [18] creatively utilized the particle swarm algorithm to optimize the parameters of CNN, thereby enhancing the model's generalization performance and fast learning ability. Additionally, autoencoders (AEs), generative adversarial networks (GANs), and capsule networks (CapsNets) have also been widely applied in HSI classification. However, the networks above face the following urgent issues that need to be addressed.

1) RNNs exhibit high computational complexity during training and inference, limiting their scalability in handling long sequences and large-scale datasets.
2) As the sequence length increases, the computational complexity of Transformers grows quadratically, creating a bottleneck that limits their efficiency.
3) The weights of CNN kernels are fixed and cannot adaptively capture changes in spatial structure.

In recent years, benefiting from the development of state space models (SSMs) [19], which alleviates the problem of low computational efficiency in RNNs and Transformers, a new perspective for modeling sequence information has been provided. Additionally, the computational complexity and memory usage of SSM are linearly related to the sequence length, which is crucial for enabling fast training and inference. Furthermore, to address the long-range dependency modeling issue in limited storage space with SSM, Gu et al. [20] introduced HiPPO, which compresses input signals into coefficient vectors and constructs a new memory update mechanism to retain all history. For performing inference calculations on ultra-long sequences, the structured state-space sequence model (S4) Gu et al. [21] was proposed, further simplifying SSM to the computation of a Cauchy kernel. Recently, a new SSM-based framework called Mamba [22] has been introduced. In language and audio tasks [23], Mamba has been shown to surpass Transformers of equivalent scale. Its selective information processing, simpler design architecture, and parallel scanning algorithm make it more efficient, with broad application prospects. The rapid development of SSM has revealed its enormous potential and high adaptability in sequence data mining. However, the high-dimensional spectral data and information redundancy pose challenges for the application of SSM in HSI classification.

Graph convolutional networks (GCNs) [24] are more flexible than CNNs in capturing dynamic spatial structures. GCNs can extract spatial features from topological graphs of non-Euclidean geometric structures and explicitly aggregate node features based on the similarity between pixels to construct adjacency matrices [25]. To help GCNs better understand global contextual information, Ding et al. [26] employed a multiscale graph aggregation network, graphSAGE, to automatically learn deep semantic information of HSI. Xu et al. [27] optimized GCN using fuzzy theory to build robust graphs for better representing complex spatial–spectral relationships in HSI. To better capture pixel-level features and alleviate the overfitting issue of GCN, Yu et al. [28] proposed a novel two-branch deeper GCN (TBDGCN), which employs the DropEdge technique and residual connections to mitigate the overfitting problem. Su et al. [29] developed a graph transmission network based on adaptive cuckoo refinement (ACGT-Net), which enhances the generalization performance of GCN through the integration of the cuckoo search strategy (CSS).

To address the challenges above and achieve efficient parallel computation while adaptively mining spatial feature relationships, we explore the potential of SSM in HSI classification. We propose an efficient graph structure learning vision mamba for HSI classification—GraphMamba. GraphMamba incorporates a series of optimized designs tailored to the characteristics of HSI and serves as an end-to-end efficient joint feature extraction and classification network. Specifically, to reduce training resource consumption for efficient computation, GraphMamba utilizes spatial–spectral information cubes as input, preserving local spatial–spectral characteristics while alleviating computational pressure. In addition, we introduce a global mask (GM) to refine high-dimensional data and propose HyperMamba to extract deep semantic information. Finally, we design SpatialGCN to implement spatial graph structure learning, employing a multiorder neighborhood aggregation matrix to capture global graph structure

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information better. The main contributions of this article are as follows.

1) We introduce a novel graph structure learning vision Mamba paradigm, which achieves efficient representation of deep spatial–spectral features through the adaptive GraphMamba encoder.
2) We propose HyperMamba, which reduces information redundancy interference by introducing a global spectral mask and decreases computational costs through parallel scanning.
3) We design SpatialGCN with enhanced context-awareness capabilities, utilizing a multihop adaptive neighborhood matrix to mine dynamic spatial structural features better.

II. Review of SSM and GCN

A. State Space Models

The SSM is a mathematical model that effectively manages sequential data in deep learning. The SSM maps the input vector $x(t) \in \mathbb{R}^m$ and the state vector $h(t) \in \mathbb{R}^n$ to the output vector $y(t) \in \mathbb{R}^p$. Deep learning networks derived from SSM can overcome the issue of Transformer being insensitive to discrete features in sequence prediction tasks, thereby establishing stronger logic for temporal data.

The key components of SSM are the state representation equation and the prediction output equation. The continuous SSM representation is as follows:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times m}$, and $\mathbf{C} \in \mathbb{R}^{p\times n}$ represent the state transition matrix, input matrix, and output matrix, respectively, which are fixed and unchanged. $h'(t) \in \mathbb{R}^n$ is the derivative of state vector $h(t) \in \mathbb{R}^n$ at time t.

In image processing tasks, signals are often discrete rather than continuous. To better handle discrete sequences, continuous SSM needs to be transformed into discrete SSM

$$h_k = \overline{\mathbf{A}}h_{k-1} + \overline{\mathbf{B}}x_k \tag{3}$$

$$y_k = \overline{\mathbf{C}}h_k \tag{4}$$

$$\overline{\mathbf{A}} = e^{\Delta \mathrm{A}} \tag{5}$$

$$\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1} \cdot \left(e^{\Delta\mathbf{A}} - \mathbf{I}\right) \cdot \Delta\mathbf{B} \tag{6}$$

$$\overline{\mathbf{C}} = \mathbf{C} \tag{7}$$

where k represents the step size and Δ is the sampling period. $\overline{\mathbf{A}} \in \mathbb{R}^{n\times n}$, $\overline{\mathbf{B}} \in \mathbb{R}^{n\times m}$, and $\overline{\mathbf{C}} \in \mathbb{R}^{p\times n}$ represent the discrete matrices obtained through the zero-order hold technique. $\mathbf{I} \in \mathbb{R}^{n\times n}$ is the identity matrix. $x_k \in \mathbb{R}^m$, $h_k \in \mathbb{R}^n$, and $y_k \in \mathbb{R}^p$ represent the discrete input sequence, discrete latent state representation, and discrete predicted output sequence, respectively.

B. Graph Convolutional Network

GCNs can represent data as undirected graphs, establishing spatial relationships between different land covers. For a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, $\mathcal{V}$ denotes a set of vertices $|\mathcal{V}| = \mathcal{N}$, $\mathcal{E}$ is the edge set, $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the adjacency matrix of $\mathcal{G}$. Given the adjacency matrix $\mathbf{A}$, the Laplacian matrix $\mathbf{L} \in \mathbb{R}^{N\times N}$ of the $\mathcal{G}$ is defined as

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{8}$$

where $\mathbf{D} \in \mathbb{R}^{N\times N}$ is the degree matrix of $\mathcal{G}$.

Normalizing the Laplace matrix $\mathbf{L}$ yields

$$\mathbf{L}_n = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} \tag{9}$$

where $\mathbf{I} \in \mathbb{R}^{N\times N}$ is the identity matrix and $\mathbf{L}_n \in \mathbb{R}^{N\times N}$ is the normalized Laplacian matrix.

GCN introduces a convolutional kernel $g_\theta = \mathrm{diag}(\theta)$ to embed nodes, where $\theta \in R^N$. The formula for graph convolution can be expressed as

$$x_G * g_\theta = \mathbf{U}g_\theta(\Lambda)\mathbf{U}^T x_G \tag{10}$$

where $x_G \in \mathbb{R}^{N\times D}$ is the node representation, $\mathbf{U} \in \mathbb{R}^{N\times N}$ is the eigenvector matrix of $\mathbf{L}_n$, and Λ is the eigenvalues of $\mathbf{L}_n$. To simplify the calculation, the Chebyshev polynomial is used to fit $g_\theta(\Lambda)$

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K} \theta_K T_K(\mathbf{L}_n) \tag{11}$$

where T_K is the Chebyshev polynomial.

Taking the first-order Chebyshev polynomial for further simplification, we can obtain

$$x_G * g_\theta = \theta\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\right)x_G \tag{12}$$

where $\tilde{A} = \mathbf{A} + \mathbf{I}$, and $\tilde{D} = \sum_j \tilde{A}_{ij}$.

Based on (12), the propagation rule of GCN is as follows:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{13}$$

where $\mathbf{H}^{(l+1)} \in \mathbb{R}^{N\times D'}$ and $\mathbf{H}^{(l)} \in \mathbb{R}^{N\times D'}$ represent the features of the $l+1$ and l layers respectively, $\sigma(\cdot)$ represents the activation function, and $\mathbf{W} \in \mathbb{R}^{D'\times D'}$ represents the weight matrix.

III. Proposed Methodology

In this section, we first introduce a novel HSI processing paradigm, which establishes spatial–spectral cubes to retain local information while reducing the input volume through block-wise input. Next, to address the issues of spectral redundancy and high-computational resource consumption, we design the HyperMamba module to introduce GM for selective extraction of spectral features and parallel computation. Additionally, we introduce the SpectialGCN module to achieve adaptive spatial information mining by aggregating neighborhood information through WMA. Finally, an analysis of model complexity is conducted.

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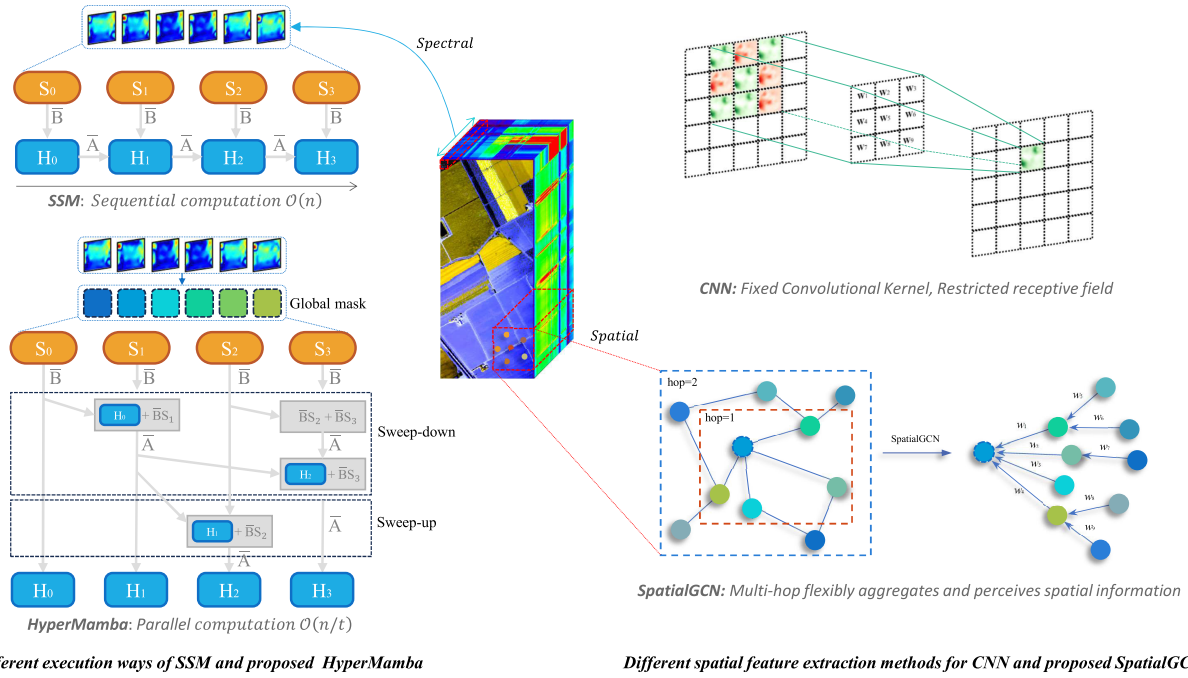


Fig. 1. Improvements of the proposed spatial–spectral feature extraction modules HyperMamba and SpatialGCN compared with other frameworks are discussed in this article.

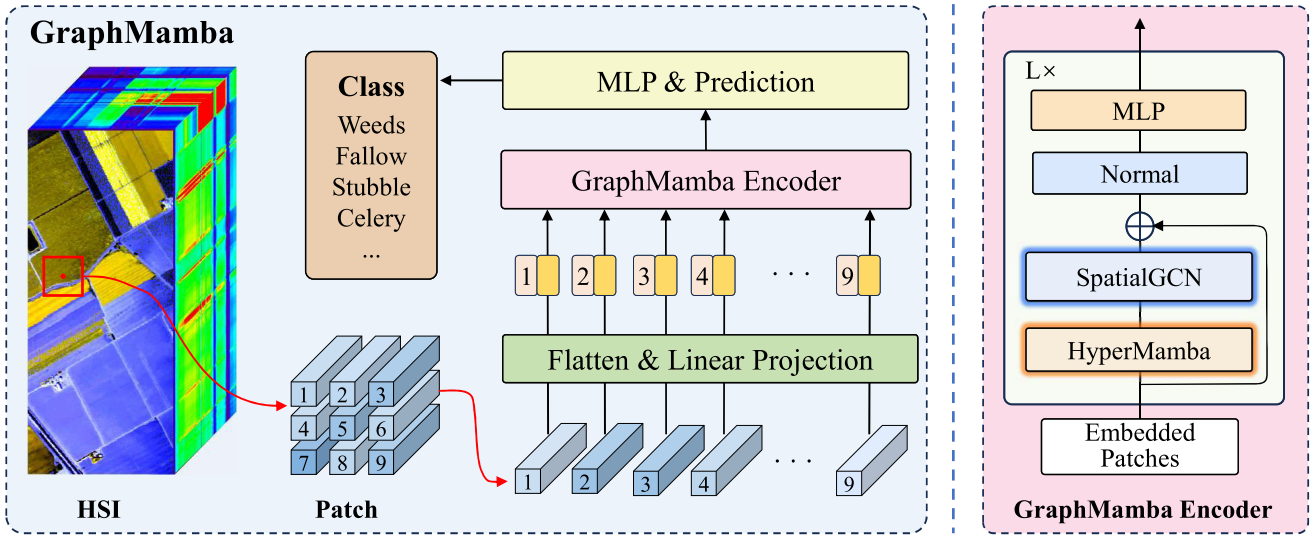


Fig. 2. Overall view of GraphMamba. We first segment HSI into multiple spatial–spectral cubes, then linearly project them into patch tokens, and then send the token sequences to the proposed GraphMamba encoders to extract features. Finally, classification prediction features are obtained through MLP.

A. GraphMamba

The HSI distinguishes land cover types by differences in pixel values (reflecting the spectral information) and spatial variations (indicating the spatial information). Compared with RNN and Transformer, Mamba utilizes SSM to dynamically filter and process information based on inputs, allowing the model to remember or ignore input parts selectively, thus enabling more effective processing of spectral sequences, as shown in Fig. 1.

This article aims to develop a universal and scalable HSI classification baseline network based on Mamba. To achieve this, we propose a novel hyperspectral information processing paradigm by partitioning HSI into spatial–spectral cubes to retain spectral information while reducing computational burdens. The overall architecture of GraphMamba is illustrated in Fig. 2.

Mamba usually encodes 1-D sequences as tokens. To facilitate easier processing by the feature extractor, we unfold the 3-D image $\mathbf{h} \in \mathbb{R}^{H\times W\times D}$ into N 2-D patches $\mathbf{x} \in \mathbb{R}^{N\times(P^2\cdot D)}$ in the form of Follow Patch [12], where H, W and D represent the length, width, and spectral dimension of the hyperspectral image, P is the patch size, and N represents the number of patches. Then, we linearly project $\mathbf{x}$ into C dimensions and

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introduce position vector $\mathbf{F}_{\text{pos}} \in \mathbb{R}^{N\times C}$

$$\mathbf{M}_0 = \left[\mathbf{x}_p^1\mathbf{W}; \mathbf{x}_p^2\mathbf{W}; \cdots \mathbf{x}_p^n\mathbf{W}; \cdots \mathbf{x}_p^N\mathbf{W}\right] + \mathbf{F}_{\text{pos}} \tag{14}$$

where $\mathbf{x}_p^n \in \mathbb{R}^{P^2\cdot D}$ denotes the n-th patch of $\mathbf{x}$, $\mathbf{W} \in \mathbb{R}^{(P^2\cdot D)\times C}$ is a learnable parameter matrix, and $\mathbf{M}_0 \in \mathbb{R}^{N\times(P^2\cdot C)}$ is the output feature after linear processing. Unlike ViT [30], we do not use a class token to represent the category prediction feature of the entire patch sequence. Since HSI classification is performed for each pixel, we extract the classified pixel feature $\mathbf{x}_p^c \in \mathbb{R}^{1\times C}$ from each patch as output features.

We input the features $\mathbf{M}_l \in \mathbb{R}^{N\times(P^2\cdot C)}$ obtained from the linear projection into the l-th encoding layer GrM$(\cdot)$ of GraphMamba to obtain the output $\mathbf{M}_{l+1} \in \mathbb{R}^{N\times(P^2\cdot C)}$. Then, we normalize the classified pixel features $\mathbf{M}_{l+1}^c \in \mathbb{R}^{N\times 1\times C}$ and feed them into a multilayer perceptron to obtain the final prediction vector $y \in \mathbb{R}^{N\times C}$. The process is as follows:

$$\mathbf{M}_{l+1} = \text{GrM}(\mathbf{M}_l) + \mathbf{M}_l \tag{15}$$

$$\mathbf{M'}_c = \text{Norm}\left(\mathbf{M}_{l+1}^c\right) \tag{16}$$

$$\mathbf{y} = \text{MLP}\left(\mathbf{M'}_c\right) \tag{17}$$

where $\mathbf{M'}_c \in \mathbb{R}^{N\times C}$ represents the normalized output features of the classified pixel features $\mathbf{M}_{l+1}^c$. Norm$(\cdot)$ and MLP$(\cdot)$ denote the regularization layer and the multilayer perceptron, respectively. We do not use a class token; instead, we directly use the classified pixel features for classification. The advantage of this approach is that the extracted features are more targeted, which also helps to simplify the network structure further.

B. HyperMamba

HSIs have high spectral dimensions and large data volume [31]. To efficiently establish a bridge for information exchange between different bands, focusing on more discriminative information, this article proposes the GM to aggregate spectral information based on the correlation between bands, while alleviating the issue of information redundancy. Assuming the input features are denoted by $\mathbf{M} \in \mathbb{R}^{N\times C}$, the computation process of the GM is as follows:

$$\hat{\mathbf{M}} = \text{Conv}\left(\mathbf{R} \times \mathbf{S}^{\text{T}}\right)\mathbf{M} \tag{18}$$

$$\mathbf{R} = \mathbf{W}_R \cdot \mathbf{M} \tag{19}$$

$$\mathbf{S} = \mathbf{W}_S \cdot \mathbf{M} \tag{20}$$

where "$\cdot$" and "$\times$" represent dot product and matrix multiplication, respectively. $\mathbf{W}_R, \mathbf{W}_C \in \mathbb{R}^{N\times C}$ denote learnable weight parameters, while Conv$(\cdot)$ represents 2-D convolution. $\mathbf{R}, \mathbf{S} \in \mathbb{R}^{N\times C}$ are the features after linear processing. $\mathbf{R}$, and $\mathbf{S}$ form a GM $\mathbf{U}$ through convolution Conv$(\mathbf{R} \times \mathbf{S}^{\text{T}})$, and obtain the output $\hat{\mathbf{M}}$. This process utilizes a spectral domain attention mechanism to reduce the impact of redundant information and extract discriminative spectral information.

Residual connections are often used between different encoding blocks to promote information exchange and gradient propagation between networks, which can enhance the model's expressive power and feature utilization [32]. However, directly using residual connections for the HSI classification model can easily introduce original spectral noise [33]. To reduce the model's sensitivity to noise, this article designs an automatic residual (AutoRes) connection to selectively incorporate original information, as shown in Fig. 3. Assuming the input features of the lth encoding layer are denoted by $\mathbf{M}_l \in \mathbb{R}^{N\times(P^2\cdot C)}$, AutoRes is represented as follows:

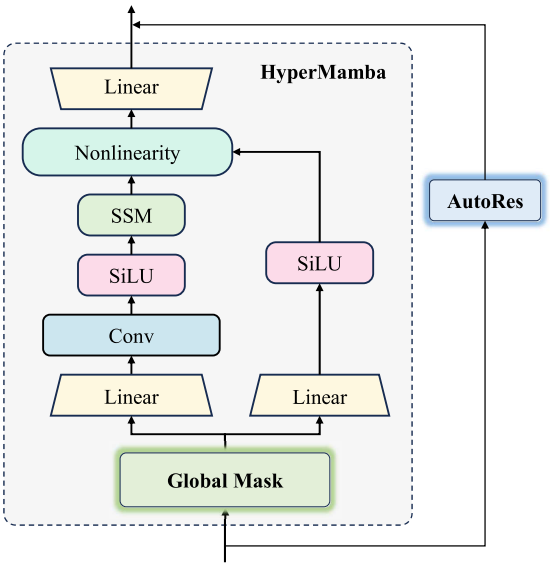


Fig. 3. Diagram of the HyperMamba in the proposed GraphMamba. The luminous marking modules are GM and AutoRes, respectively.

$$\begin{aligned}\mathbf{Z} &= \text{AutoRes}(\mathbf{M}_l, \mathbf{M}_{l-1}) \\ &= \varepsilon\mathbf{M}_l + (1-\varepsilon)\mathbf{M}_{l-1}\end{aligned} \tag{21}$$

where ε is a learnable parameter, and $\mathbf{Z} \in \mathbb{R}^{N\times(P^2\cdot C)}$ represents the output features. The AutoRes module is concise and efficient, as it adaptively integrates output features and original information, alleviating the issue of over-smoothing features.

C. SpatialGCN

GCN possesses strong spatial information aggregation capabilities and flexibility. However, the graphs are typically manually constructed, and the graphs used often struggle to represent global contextual information [34]. Existing solutions often involve constructing multiple graph branches with different neighborhood ranges to expand the perceptual scope [35] [36]. However, this approach not only leads to a quadratic increase in computational complexity but also tends to introduce aberrant information [37].

To adaptively capture spatial contextual information and further enhance the model's generalization ability, this article proposes weighted multihop aggregation (WMA) Spatial Graph Convolutional Encoding (SpatialGCN). Assuming the input is $\mathbf{Z} \in \mathbb{R}^{N\times C}$, SpatialGCN can be represented as

$$\mathbf{H}^p = \varphi\left(\sum_{n=0}^{p}\left(\mathbf{Q}^n \times \mathbf{Z}^n\right)\mathbf{W}\right) \tag{22}$$

$$\mathbf{Z}^0 = \mathbf{Z} \tag{23}$$

$$\mathbf{Z}^p = \tilde{\mathbf{D}}_p^{\frac{1}{2}}\tilde{A}_p\tilde{\mathbf{D}}_p^{-\frac{1}{2}}\mathbf{Z} \tag{24}$$

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Algorithm 1 Mechanism of GraphMamba

Input: Hyperspectral image **h, learning rate** η**, number of epochs** $\mathcal{T}$**, patch size** P**;**

Divide the Hyperspectral image **h into** $\mathcal{N}$ **patches of size** $P^2 \cdot D$ **denoted as** $\mathbf{x}_p$**;**

// Train GraphMamba model.

for $\tau = 1$ **to** $\mathcal{T}$ **do**

  for $n = 1$ **to** $\mathcal{N}$ **do**

    // Patch flattening and linear processing.

    Linearly project $\mathbf{x}_p$ into C dimensions, and introduce the position vector $\mathbf{F}_{pos}$ to obtain the output features $\mathbf{M}_0$ of the linear processing layer, according to Eq.14;

    // HyperMamba extracts spectral features.

    Calculate the global mask $\mathbf{U}$ according to Eq.18, and use the global mask $\mathbf{U}$ to filter the spectral information of $\mathbf{M}_0$ to obtain the output $\hat{\mathbf{M}}$;

    The feature $\hat{\mathbf{M}}$ is transformed into a more efficient and discriminative feature representation through the HyperMamba module, and the output $\mathbf{Z}$ is obtained according to Eq. 21 through automatic residual connection;

    // SpatialGCN extracts spatial contextual information.

    Construct an adaptive filtering matrix $\mathbf{Q}^n$ according to Eq.25, and capture the spatial contextual information of $\mathbf{Z}$ through SpatialGCN to obtain output $\mathbf{H}^p$ according to Eq.22;

    $\mathbf{H}^p$ is processed through Normal and MLP modules to obtain final output feature **y;**

    Calculate the loss based on the cross-entropy loss function and update the network parameters using Adam optimizer;

Use the trained network to predict the test samples.

Output: Predict all pixel land cover classes L.

where $\mathbf{Q}^n \in \mathbb{R}^{N\times N}$ represents the adaptive filtering matrix, p represents the p-hop neighboring nodes of the node, $\mathbf{Z}^p \in \mathbb{R}^{N\times C}$ denotes the p-hop aggregated features of the node, and when p=0, $\mathbf{Z}^0$=$\mathbf{Z}$. $\mathbf{A}_p, \mathbf{D}_p \in \mathbb{R}^{N\times N}$ respectively represent the adjacency matrix and the degree matrix of n-hop connections. $\varphi(\cdot)$ denotes the activation function, and $\mathbf{W} \in \mathbb{R}^{C\times C'}$ is the learnable shared parameter matrix. When p=1, $\mathbf{H}^p \in \mathbb{R}^{N\times C'}$ is a normal GCN with residual connections, providing good scalability and adaptability.

Additionally, the calculation of the adaptive filtering matrix $\mathbf{Q}^n$ is as follows:

$$q_{ij} = \begin{cases} \dfrac{\exp\left(-\gamma \|s_i - s_j\|^2\right)}{\sum_{j=0}^{n} \exp\left(-\gamma \|s_i - s_j\|^2\right)}, & if\ a_{ij}^p \neq 0 \\ 0, & if\ a_{ij}^p = 0 \end{cases} \tag{25}$$

where q_{ij} represents the element in the matrix $\mathbf{Q}^n$, s_i denotes the i-th node in the graph, $\gamma = 0.2$ is an empirical value [38], and a_{ij}^p is the value at position (i, j) in the adjacency matrix $\mathbf{A_p}$. Different from the simple way of adding branches in a multihop GCN, SpatialGCN adopts weight sharing to reduce the parameter volume, thereby alleviating computational pressure effectively. In addition, adaptive filtering matrices can abstract the relevance of different nodes and better extract spatial structural features. We provide the pseudocode for GraphMamba in Algorithm 1.

D. Computational Complexity Analysis

Given a sequence of n spectral features $\mathbf{h} \in \mathbb{R}^{p\times d}$, where p represents the product of the height and width of a patch and d is the dimension of the features after linear processing. The computational complexity of GM is $\mathcal{O}(2np^2d)$, the computational complexity of the SSM module is $\mathcal{O}(3p(2d)n + p(2d)n)$, and the computational complexity of GCN is $\mathcal{O}(|\mathcal{E}|p^2d)$, where $|\mathcal{E}|$ denotes the number of nonzero entries in the adjacency matrix $\mathbf{A}$. It can be observed that the computational complexity of SSM grows linearly with the sequence length and feature dimension, greatly enhancing computational efficiency.

IV. Experiment

In this section, we conducted comprehensive experiments to validate the effectiveness of the proposed GraphMamba. First, we evaluated the algorithm's performance by testing 11 algorithms on three datasets of different sizes using three quantitative evaluation metrics. Additionally, we generated classification maps to compare the classification results qualitatively. Subsequently, we compared the advancement of the algorithms at different sampling ratios on OAs and conducted a complexity analysis of the models. Finally, we verified the efficacy of the proposed modules through ablation experiments.

A. Data Description

Three real benchmark datasets, Indian Pines (IP), Salinas (SA), and the University of Houston 2018 (UH2018), are employed to test the algorithm performance from various perspectives. The IP dataset is used to evaluate the classification ability under conditions of uneven sample distribution, the SA dataset is utilized to validate the classification ability under similar spectral conditions of different land covers, and the UH2018 dataset is employed to verify the classification performance of large-scale datasets with high spatial resolution.

The Indian Pines (IP) dataset has a size of 145 × 145, 200 effective bands, 16 land covers, and a spatial resolution of approximately 20 m.

The Salinas (SA) dataset has dimensions of 512 × 217, 204 effective bands, 16 land covers, and a spatial resolution of 3.7 m.

The UH2018 dataset contains 601 × 2385 pixels and 48 spectral bands ranging from 380 to 1050 nm. It contains 20 land covers with a spatial resolution of 1 m.

It is worth noting that we trained on a smaller scale of data, using 2%, 0.2%, and 0.2% of the labeled samples for IP, SA, and UH2018, respectively. The pseudo-color images and ground-truth maps of the dataset are described in Fig. 4. The details of the sample selection from the dataset are presented in Table I.

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TABLE I

DETAILS OF EACH LAND-COVER CLASS OF THREE DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

NO.	UH2018				Indian Pines				Salinas			
	Class name	Train	Validation	Test	Class name	Train	Validation	Test	Class name	Train	Validation	Test
1	Healthy grass	19	19	9780	Alfalfa	3	3	43	Weeds-1	4	4	2005
2	Stressed grass	65	65	32437	Corn-notill	28	28	1400	Weeds-2	7	7	3719
3	Artificial turf	3	3	681	Corn-mintill	16	16	814	Fallow	3	3	1973
4	Evergreen trees	27	27	13568	Corn	4	4	233	Fallow-rough-plow	3	3	1391
5	Deciduous trees	10	10	5011	Grass-pasture	9	9	474	Fallow-smooth	5	5	2673
6	Bare earth	9	9	4507	Grass-trees	14	14	716	Stubble	7	7	3952
7	Water	3	3	263	Grass-pasture-mowed	3	3	25	Celery	7	7	3572
8	Residential buildings	79	79	39693	Hay-windrowed	9	9	469	Grapes-untrained	22	22	11249
9	Non-residential buildings	447	447	223305	Oats	3	3	17	Soil-vinyard-develop	12	12	6191
10	Roads	91	91	45775	Soybeans-notill	19	19	953	Corn	6	6	3272
11	Sidewalks	68	68	33961	Soybean-mintill	49	49	2406	Lettuce -4wk	3	3	1065
12	Crosswalks	3	3	1515	Soybean-clean	11	11	582	Lettuce -5wk	3	3	1924
13	Major thoroughfares	92	92	46256	Wheat	4	4	201	Lettuce -6wk	3	3	913
14	Highways	19	19	9846	Woods	25	25	1240	Lettuce -7wk	3	3	1067
15	Railways	13	13	6924	Buildings-grass-trees-drivers	7	7	379	Vinyard-untrained	14	14	7254
16	Paved parking lots	23	23	11477	Stone-steel-towers	3	3	90	Vinyard -trellis	3	3	1804
17	Unpaved parking lots	3	3	143								
18	Cars	13	13	6534								
19	Trains	10	10	5359								
20	Stadium seats	13	13	6811								
Total		1010	1010	503846		207	207	10042		105	105	54024

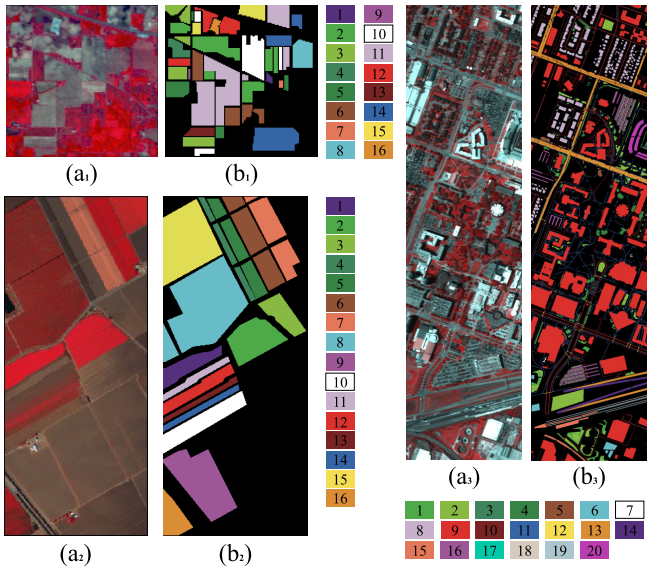


Fig. 4. Three datasets. (a1), (b1), and (c1) are the false color images of IP, SA, UH2018. (a2), (b2), and (c2) are the ground-truth maps.

B. Experiments Settings

1) Comparative Methods: To ensure the reliability of the experimental results to the fullest extent, we conducted a detailed comparison using ten baseline methods. These ten methods can be roughly categorized into traditional machine learning method such as SVM-RBF [39]; CNN-based methods including 3-D-CNN [40], DFFN [41], and RSSAN [42]; LSTM-based method such as AB-LSTM [43]; Transformer-based methods such as spectral former (SF) [33] and GAHT [11], fusion methods combining GCN and CNN such as WFCG [44], and fusion methods combining CNN and Transformer such as CACFTNet [45] and HSDBIN [46].

2) Evaluation Metrics: The overall classification accuracy (OA) is used to observe the classification accuracy of all samples. The average accuracy (AA) represents the average accuracy of different class classifications. The Kappa coefficient is employed as an indicator of the degree of consistency. Additionally, the classification accuracy of each class is specifically presented.

3) Parameter Settings and Implementation Details: To facilitate fair comparisons, all experiments were conducted on a workstation using the PyTorch platform, equipped with an Intel Xeon Silver 4210R CPU, 128 GB RAM, and an NVIDIA Titan RTX 24 GB GPU. Our proposed GraphMamba utilizes the Adam optimizer with a batch size of 64. The initial learning rate is set to 1×10^{-4} and is decayed by a factor of 0.9 after each tenth of the total epochs. We configured the number of epochs to 200 for the three datasets. For GraphMamba, we use a patch size of 11×11, and the depth of the encoder is 6 in three datasets.

C. Comparative Experimental Results and Analysis

In this section, we qualitatively and quantitatively evaluate the IP, SA, and UH2018 datasets and detail the classification results in Table II–IV, highlighting the best performance in bold. The intuitive classification results are also shown in Figs. 5–7.

1) Quantitative Analysis of Indian Pines dataset: Table II displays the comparison results of ten different methods with GraphMamba, with the best classification performance highlighted in each row. We selected various baseline methods such as CNN, LSTM, GCN, Transformer, and their improved or fused versions for comparison in the experiments to comprehensively test Mamba's HSI classification performance. Table II shows that the LSTM-based network has lower classification accuracy as it solely focuses on spectral aspects, neglecting spatial features. The fusion methods WFCG and CACFTNet, based on GCN-CNN and Transformer-CNN, have achieved good classification results due to their strong ability in spatial–spectral modeling.

Compared with other models, GraphMamba achieves the best classification performance in terms of OA, AA, and Kappa

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TABLE II

QUANTITATIVE COMPARISON OF IP DATASET. BOLD DENOTES THE BEST PERFORMANCES

Class No.	SVM-RBF	3D-CNN	DFFN	RSSAN	GAHT	AB-LSTM	SF	WFCG	CACFTNet	HSDBIN	GraphMamba
1	20.49±2.57	0.00±0.00	9.62±12.08	3.70±5.23	2.22±3.14	0.00±0.00	7.40±7.55	0.00±0.00	83.72±8.70	8.52±12.06	**88.37±0.00**
2	51.51±2.36	42.02±0.29	73.19±3.60	73.00±1.16	84.11±6.06	18.11±22.61	54.21±7.27	82.97±9.10	90.47±0.27	24.4±34.51	**93.59±0.11**
3	52.29±4.40	19.81±9.74	60.64±19.15	34.07±22.53	70.35±12.32	8.06±11.40	37.26±11.5	88.88±2.09	87.91±3.96	2.29±3.24	**96.19±0.21**
4	32.87±5.34	3.29±3.25	69.24±5.87	15.02±11.62	50.92±13.37	0.00±0.00	33.33±2.81	**98.23±1.57**	81.68±5.13	3.01±4.25	82.55±1.79
5	86.73±0.09	14.69±6.91	81.29±6.81	74.75±1.72	78.9±2.50	41.98±33.77	34.66±11.53	88.72±5.12	79.46±3.17	14.2±20.09	**91.21±0.12**
6	77.94±1.08	89.05±0.96	94.32±2.29	92.69±6.40	96.74±1.54	78.53±12.94	84.03±0.23	96.68±2.21	94.87±1.05	32.44±45.89	**98.32±0.14**
7	48.25±14.88	0.00±0.00	41.97±15.22	22.22±15.71	53.08±17.19	0.00±0.00	11.11±9.07	0.00±0.00	98.66±1.88	24.00±33.94	**100.00±0.00**
8	88.82±0.57	97.15±2.54	86.21±7.83	95.09±0.75	98.01±1.57	81.59±12.50	96.01±3.12	99.92±0.10	85.64±10.8	33.26±47.04	**100.00±0.00**
9	35.51±18.71	0.00±0.00	36.84±7.44	24.56±19.36	35.08±4.96	0.00±0.00	28.07±17.36	0.00±0.00	100.00±0.00	7.84±11.09	**100.00±0.00**
10	59.14±3.47	40.53±0.69	75.16±2.96	64.77±8.63	77.71±8.09	19.48±21.42	63.34±1.39	83.82±1.95	84.95±1.78	21.16±29.93	**92.72±0.42**
11	60.99±2.32	83.12±4.88	78.04±2.92	79.82±8.72	87.19±5.61	87.3±9.86	72.30±3.87	96.06±1.80	88.62±5.77	30.44±43.05	**97.95±0.05**
12	39.24±7.26	5.95±6.05	50.28±14.45	42.49±11.54	67.52±7.30	1.31±1.86	28.12±6.26	**93.39±3.24**	65.29±9.09	1.78±2.51	93.01±0.26
13	86.76±2.06	56.55±21.03	78.93±29.43	42.78±42.08	83.91±21.01	6.63±9.38	63.18±22.67	**97.59±1.88**	88.72±6.61	32.84±46.44	95.19±0.29
14	85.96±1.15	95.99±0.61	93.27±0.37	93.89±5.44	96.88±1.36	94.78±2.07	88.79±3.89	94.47±5.70	95.51±2.44	33.25±47.03	**100.00±0.00**
15	51.74±11.2	20.58±10.67	70.71±3.95	52.85±10.67	73.52±8.90	9.05±9.97	43.62±4.47	92.00±5.47	74.75±4.91	7.21±10.20	**96.92±0.16**
16	99.47±0.73	0.00±0.00	94.56±1.77	31.15±27.21	40.21±33.07	0.00±0.00	85.50±6.72	52.98±32.75	**99.62±0.52**	27.03±38.04	97.78±0.00
OA (%)	64.45±0.57	56.94±0.43	76.85±3.10	70.40±4.58	82.93±1.90	49.55±2.13	63.15±2.19	90.69±0.82	87.29±1.91	68.22±2.08	**95.99±0.07**
AA (%)	61.11±2.26	35.54±2.15	68.39±4.27	52.68±7.54	68.52±2.45	27.93±0.84	51.93±2.19	92.86±1.40	87.49±1.25	18.98±26.84	**95.24±0.13**
Kappa	58.96±0.58	49.13±0.59	73.61±3.58	65.96±5.11	80.44±2.17	39.61±3.78	57.85±2.38	89.40±0.93	85.51±2.13	21.13±29.87	**95.42±0.08**

TABLE III

QUANTITATIVE COMPARISON OF SA DATASET. BOLD DENOTES THE BEST PERFORMANCES

Class No.	SVM-RBF	3D-CNN	DFFN	RSSAN	GAHT	AB-LSTM	SF	WFCG	CACFTNet	HSDBIN	GraphMamba
1	**99.23±1.09**	39.90±35.37	89.06±5.20	73.35±13.22	5.93±2.41	0.05±0.07	92.44±1.93	98.35±1.67	97.15±0.85	0.00±0.00	97.85±0.94
2	98.36±0.04	99.22±0.49	97.27±1.77	99.35±0.63	99.93±0.10	65.42±46.27	98.20±1.92	100.00±0.00	99.29±0.08	33.31±47.12	**100.00±0.00**
3	79.43±10.82	14.63±12.70	62.58±8.73	62.95±13.73	67.83±16.69	0.00±0.00	57.04±12.04	88.70±5.73	59.31±2.54	20.3±28.72	**99.76±0.28**
4	97.39±0.02	93.77±2.91	87.74±5.18	96.05±1.58	98.42±0.18	0.00±0.00	95.57±0.43	96.61±1.30	96.74±1.05	33.26±47.04	**98.68±0.91**
5	84.54±2.92	96.10±1.53	87.44±5.58	93.94±4.77	94.26±2.35	12.23±17.3	83.61±1.38	97.15±0.75	96.95±0.51	33.05±46.75	**97.17±1.19**
6	99.91±0.06	99.81±0.20	98.66±1.03	98.63±0.73	100.00±0.00	85.91±14.28	98.68±0.88	99.02±0.07	**100.00±0.00**	33.30±47.09	99.24±0.03
7	93.48±2.20	**99.77±0.07**	88.4±8.51	94.97±1.38	99.02±0.71	25.07±35.45	91.84±4.55	99.50±0.50	97.41±1.37	33.24±47.02	99.30±0.05
8	68.94±4.97	82.10±6.12	79.02±0.78	85.27±8.83	88.46±4.10	**99.25±0.56**	77.33±2.89	94.96±0.78	82.17±5.31	31.06±43.94	96.50±1.65
9	95.87±2.83	99.97±0.03	95.12±1.10	98.21±1.63	99.77±0.04	97.83±1.42	97.80±0.83	**100.00±0.00**	95.64±3.88	33.33±47.14	99.95±0.09
10	79.37±10.79	41.44±16.23	70.47±13.33	86.38±1.89	82.31±10.62	23.25±20.18	72.48±13.85	92.70±2.09	71.27±3.31	22.02±31.15	**93.89±1.28**
11	86.37±11.17	0.00±0.00	75.95±21.63	39.81±20.37	82.65±9.13	28.83±22.03	59.26±28.14	95.83±3.29	95.68±1.68	29.67±41.96	**99.44±0.82**
12	89.27±4.50	91.11±4.72	84.03±12.00	96.00±4.60	99.65±0.49	0.00±0.00	83.85±8.33	98.30±1.60	**99.74±0.18**	31.22±44.15	97.94±1.56
13	89.04±2.60	0.00±0.00	34.1±39.88	26.52±37.51	69.95±27.09	0.00±0.00	23.28±23.28	80.85±2.14	**99.70±0.05**	32.31±45.69	88.94±3.92
14	92.71±3.39	16.82±6.67	93.16±5.86	61.36±24.25	97.91±1.02	0.00±0.00	96.35±2.26	98.25±1.23	97.84±1.82	31.02±43.87	**99.53±0.48**
15	63.66±6.48	38.98±17.17	70.54±4.25	38.17±26.86	72.03±5.09	0.00±0.00	72.73±5.19	91.78±4.89	74.95±7.14	17.78±25.16	**93.39±0.73**
16	98.55±0.94	29.51±9.51	68.88±9.52	78.49±8.62	86.16±5.17	0.00±0.00	71.56±9.53	**100.00±0.00**	88.74±1.13	0.18±0.26	96.14±2.07
OA (%)	83.30±1.06	70.76±1.77	82.23±1.14	80.32±2.29	89.65±0.78	46.90±6.79	82.46±1.06	96.09±0.34	88.12±0.81	78.27±2.87	**97.23±0.44**
AA (%)	88.51±0.55	58.94±1.64	80.15±1.67	76.84±3.47	89.64±1.62	27.36±5.58	79.50±1.49	95.74±0.60	90.79±0.39	25.94±36.69	**97.36±0.35**
Kappa	81.33±1.26	66.97±2.05	80.18±1.31	77.91±2.71	88.45±0.88	36.56±9.58	80.46±1.17	95.65±0.38	86.77±0.86	26.24±37.11	**96.92±0.49**

TABLE IV

QUANTITATIVE COMPARISON OF UH2018 DATASET. BOLD DENOTES THE BEST PERFORMANCES

Class No.	SVM-RBF	3D-CNN	DFFN	RSSAN	GAHT	AB-LSTM	SF	WFCG	CACFTNet	HSDBIN	GraphMamba
1	73.12±6.05	72.71±3.84	84.11±2.66	75.4±3.20	74.52±3.60	**84.96±4.11**	81.03±4.16	47.06±15.31	83.15±8.36	15.56±22.01	80.25±1.38
2	86.35±2.28	89.84±1.57	88.46±0.74	89.33±0.36	**90.43±1.41**	88.74±1.49	85.07±2.06	86.36±3.26	87.62±2.45	32.12±45.43	89.33±0.23
3	62.75±11.18	0.00±0.00	55.44±24.41	11.32±16.01	72.33±12.24	0.00±0.00	0.00±0.00	0.00±0.00	**100.00±0.00**	30.98±43.82	92.61±7.88
4	84.18±3.75	85.48±4.98	89.82±1.91	91.36±0.8	91.06±3.04	89.15±2.60	86.21±4.46	90.15±1.01	87.44±1.67	32.13±45.44	**92.71±1.65**
5	47.56±5.74	3.80±3.13	60.59±3.88	47.70±6.09	55.96±8.23	40.12±9.61	31.86±5.05	0.14±0.20	63.73±13.52	6.83±9.65	**70.47±12.36**
6	56.02±4.12	0.00±0.00	80.31±5.58	75.55±2.69	82.93±1.23	30.96±9.02	67.09±3.21	72.74±21.31	82.26±2.37	10.14±14.34	**90.04±3.02**
7	81.87±20.03	0.00±0.00	47.92±23.58	20.75±27.5	24.03±21.82	0.00±0.00	22.52±28.38	0.00±0.00	94.30±1.12	12.67±17.92	**98.10±1.37**
8	72.10±4.06	75.65±2.29	78.39±1.98	75.17±3.93	77.53±2.81	72.63±1.68	70.14±2.05	**94.51±0.28**	74.91±2.98	27.42±38.77	81.55±3.24
9	87.10±0.86	91.36±0.77	93.38±0.28	91.71±1.78	94.65±0.48	89.26±0.66	87.56±0.93	**97.00±0.61**	95.03±0.70	32.32±45.71	96.42±0.05
10	47.34±1.72	12.90±4.40	54.78±0.46	51.43±3.88	54.12±1.26	38.26±3.85	34.73±2.96	56.85±3.14	52.35±3.29	20.38±28.82	**68.45±1.36**
11	53.69±2.27	16.35±3.08	46.98±2.49	52.01±7.16	53.54±0.79	37.62±2.65	32.14±2.09	50.22±5.03	51.79±2.65	15.07±21.31	**59.02±5.93**
12	**4.98±5.72**	0.00±0.00	1.87±1.56	0.77±0.68	2.93±1.49	0.00±0.00	1.03±1.19	0.00±0.00	2.77±0.61	0.00±0.00	3.61±0.66
13	58.47±2.57	45.00±5.40	65.95±3.28	64.47±4.94	69.08±1.41	51.72±3.93	56.04±3.41	**83.67±1.54**	62.25±2.43	25.97±36.72	80.37±1.05
14	72.04±6.52	0.00±0.00	70.51±5.23	65.79±3.47	68.43±2.11	51.17±5.35	54.53±3.54	74.81±7.83	81.21±3.32	4.74±6.71	**84.08±1.84**
15	80.72±3.59	0.00±0.00	89.58±4.23	80.67±8.94	92.88±0.84	77.24±8.45	81.07±8.23	72.15±9.86	**98.55±0.25**	1.77±2.51	98.47±0.38
16	66.77±5.49	0.00±0.00	75.07±5.56	67.65±7.86	81.35±3.15	36.81±10.27	56.52±2.46	81.63±3.40	81.75±4.99	8.99±12.72	**90.33±2.17**
17	26.83±0.15	0.00±0.00	28.28±32.88	10.57±10.46	18.85±10.46	0.00±0.00	10.34±7.94	0.00±0.00	**99.77±0.33**	0.00±0.00	89.74±15.4
18	51.93±7.51	0.00±0.00	48.26±7.32	27.68±6.57	62.50±1.58	19.82±4.44	28.95±6.81	36.79±22.31	71.27±2.26	0.20±0.28	**75.09±8.04**
19	56.50±14.31	1.14±1.61	63.38±7.08	51.44±10.15	59.76±6.06	24.15±8.16	42.64±10.47	66.33±3.19	48.79±4.82	0.00±0.00	**81.85±7.66**
20	82.92±13.73	0.00±0.01	77.79±9.32	76.34±2.95	81.85±4.25	52.76±27.53	70.5±10.22	**98.21±2.09**	71.93±1.12	0.05±0.08	95.58±2.56
OA (%)	74.73±1.27	62.40±0.30	79.61±0.37	77.39±1.35	81.08±0.31	70.60±1.46	70.35±0.70	83.04±0.77	80.39±0.59	76.59±0.16	**86.45±0.13**
AA (%)	62.66±0.44	24.71±0.37	65.04±4.99	56.35±2.79	65.43±2.87	44.26±2.08	49.99±2.29	55.43±2.17	74.54±1.40	13.87±19.61	**80.90±0.77**
Kappa	66.98±1.59	46.59±0.38	73.38±0.54	70.55±1.62	75.23±0.42	61.65±1.94	61.37±0.77	77.71±1.05	74.31±0.82	22.51±31.83	**82.29±0.18**

metrics, with improvements of 5.30%, 2.38%, and 6.02%, respectively, over the second-best performing model. This demonstrates the effectiveness of the proposed graph-based learning approach in Mamba.

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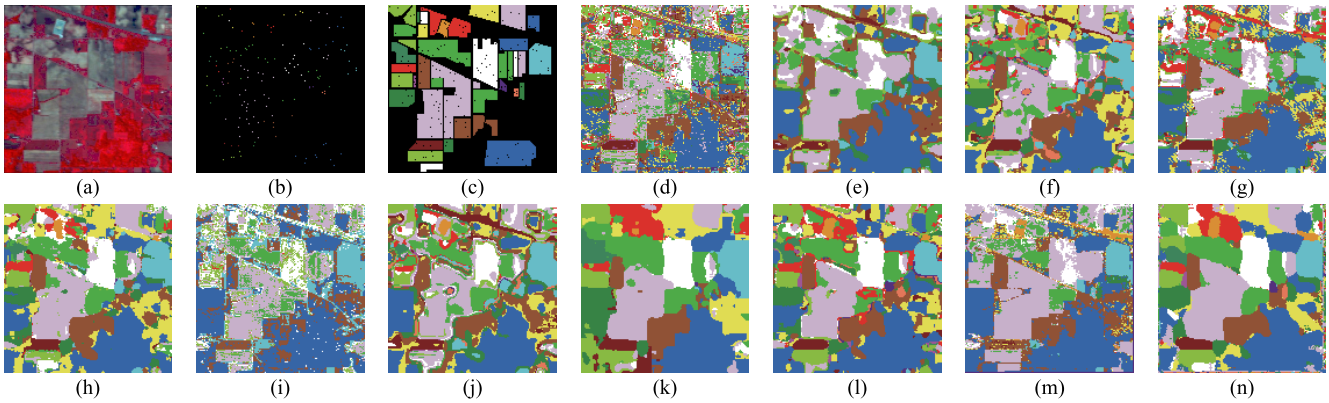


Fig. 5. Classification maps of IP. (a) Pseudo-color image. (b) Train labels. (c) Test labels. (d) SVM-RBF. (e) 3-D-CNN. (f) DFFN. (g) RSSAN. (h) GAHT. (i) AB-LSTM. (j) SF. (k) WFCG. (l) CACFTNet. (m) HSDBIN. (n) GraphMamba.

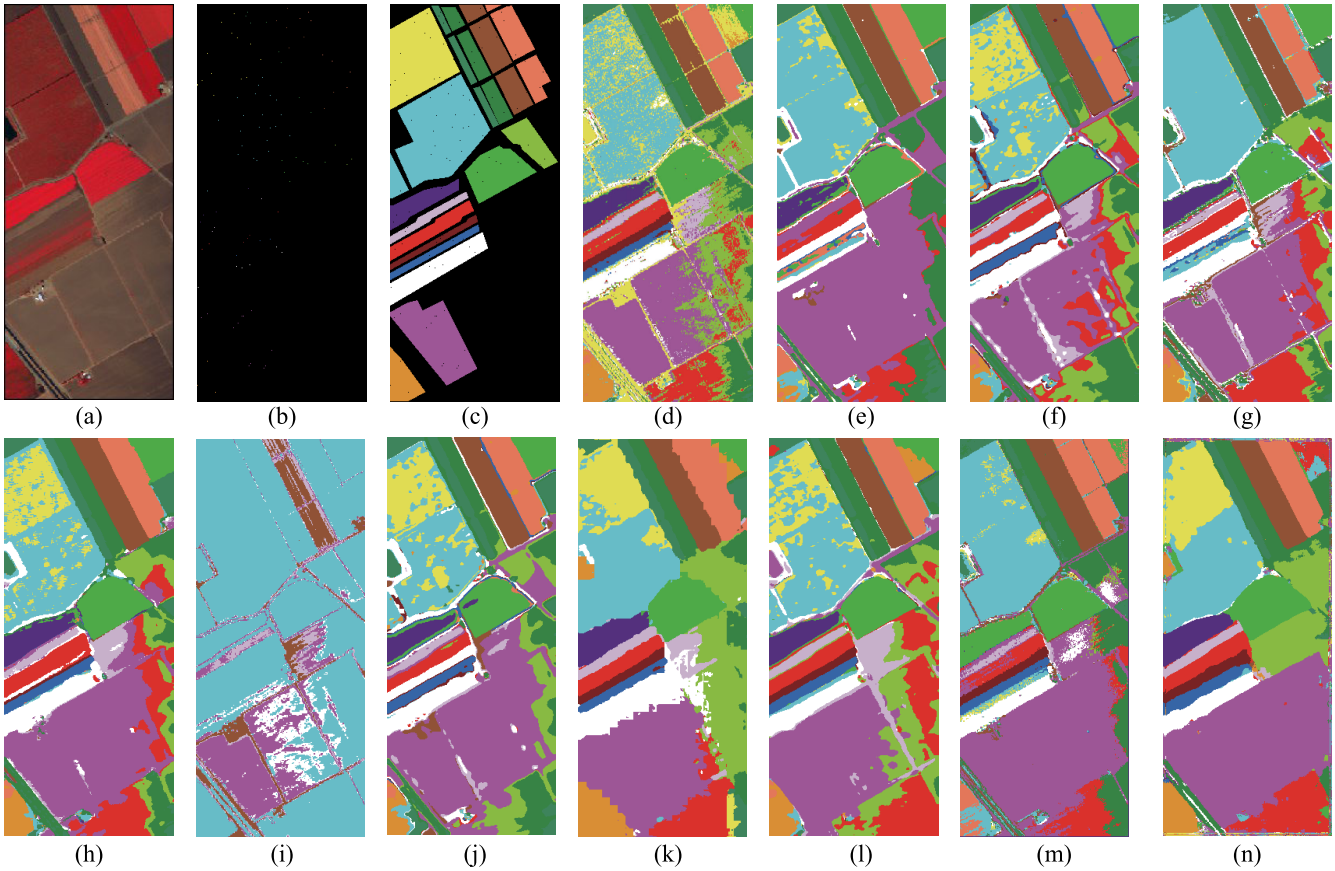


Fig. 6. Classification maps of SA. (a) Pseudo-color image. (b) Train labels. (c) Test labels. (d) SVM-RBF. (e) 3-D-CNN. (f) DFFN. (g) RSSAN. (h) GAHT. (i) AB-LSTM. (j) SF. (k) WFCG. (l) CACFTNet. (m) HSDBIN. (n) GraphMamba.

Regarding visualization results, traditional methods like SVM-RBF tend to exhibit salt-and-pepper noise, while deep learning-based classification methods produce smoother classification maps. Furthermore, our method shows better handling of edge features, significantly reducing edge noise. This is attributed to the context-aware GCN encoding module, which better aggregates spatial information and reduces instances of misclassification. Additionally, the HyperMamba encoder enhances the modeling of sequences, reducing noise during the classification process.

2) Quantitative Analysis of Salinas dataset: Similar terrain features in the SA dataset pose a challenge for classification. In this scenario, GraphorMamba continues to exhibit the most dominant classification results, with improvements of 1.14%, 1.62%, and 1.27% in OA, AA, and Kappa, respectively, compared with the second-best classification method. Furthermore, the classification results of our approach have a small standard deviation, indicating a high level of algorithmic stability. Due to the spectral similarity between Corn and Vinyard untrained land covers, their classification accuracy is relatively low.

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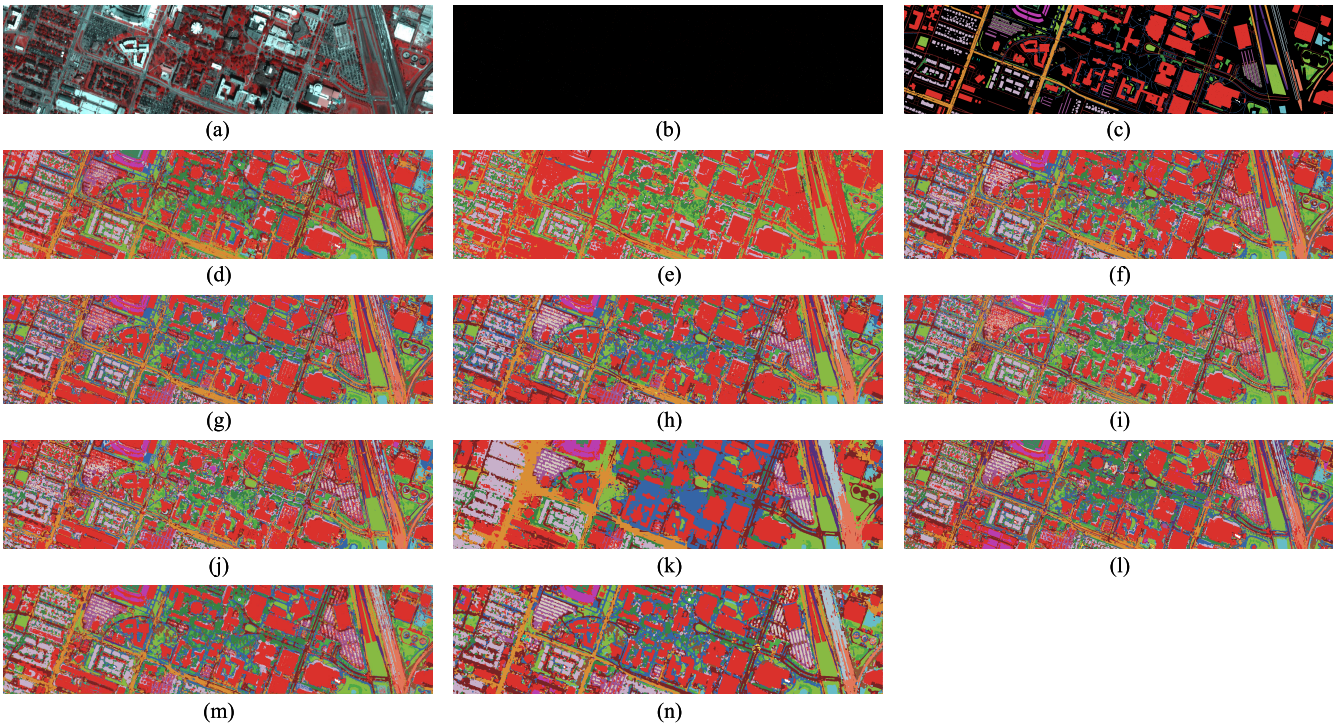


Fig. 7. Classification maps of UH2018. (a) Pseudo-color image. (b) Train Labels. (c) Test labels. (d) SVM-RBF. (e) 3-D-CNN. (f) DFFN. (g) RSSAN. (h) GAHT. (i) AB-LSTM. (j) SF. (k) WFCG. (l) CACFTNet. (m) HSDBIN. (n) GraphMamba.

However, GraphMamba achieves superior classification results among all methods by integrating spatial–spectral features more effectively through a multiscale contextual attention mechanism.

The classification maps show that GraphMamba delivers satisfactory classification results even with a small proportion of training data. Given the proximity of terrain features, misclassifications, and noise are prone to occur at the edges. GraphMamba effectively addresses this issue, resulting in a smoother visual outcome. At the edges, misclassification and noise are prone. Graphmamba effectively addresses this issue, creating clear boundaries for land features and achieving a smoother visual effect.

3) Quantitative Analysis of UH2018 dataset: For the large-scale dataset UH2018, its larger size and more dispersed land cover types pose significant challenges for accurate classification. Due to the limited number of training samples and the high classification difficulty, the overall accuracy of these classification algorithms on the UH2018 dataset is lower than that on the IP and SA datasets. GraphMmaba utilized 1010 pixel points (accounting for 0.2% of the total pixel points), achieving an overall accuracy of 86.45%. Compared with the second-best classification performance, its OA improved by 3.41%. Additionally, the smaller standard deviation indicates the stability of its algorithm.

The natural advantage of Mamba in handling sequences within HyperMamba, combined with the proposed GM module, enhances its ability to capture spectral information. In contrast to CNNs, the SpatialGCN module can more flexibly process spatial information and is also effective in the context of large-scale heterogeneous objects.

From Fig. 7, it can be seen that even for smaller objects, GraphMamba can achieve accurate classification, while other methods are prone to misclassifying smaller objects entirely. GraphMamba's texture handling is also quite satisfactory, showing a high degree of distinction between different categories. This indicates that integrating spatial and spectral information for feature modeling can better represent object relationships, enhancing classification performance.

D. Analysis of Training Samples

Different sampling ratios may lead to performance differences. To evaluate the adaptability of GraphMamba under different proportions of training samples, experiments were conducted using five different sampling ratios across ten methods. Specifically, for the IP dataset, training samples are taken at 1%, 2%, 3%, 4%, and 5% of the dataset, while for the SA and UH2018 datasets, training samples are taken at 0.25%, 0.5%, 1%, 2%, and 3%. Fig. 8 illustrates how the OA varies with the number of samples on the three datasets. As the number of samples increases, the OA also increases. It is noteworthy that the OA of GraphMamba (as indicated by the blue line) consistently remains at the highest level, demonstrating its strong robustness and generalization capability. This is attributed to GraphMamba's adaptive spatial contextual modeling and spectral feature extraction capabilities.

E. Visualization Analysis of Features

In this section, we use t-distributed stochastic neighbor embedding (t-SNE) [47] to visualize the distribution of output features from WFCG, CACFTNet, and GraphMamba across

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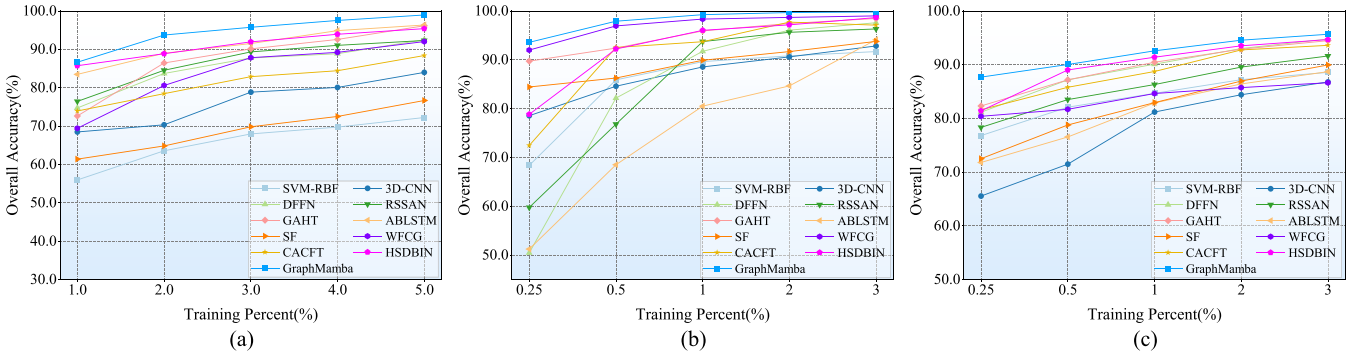


Fig. 8. OA for different proportions of training samples on the three datasets. (a)–(c) OA curves in the IP, SA, and UH2018 datasets.

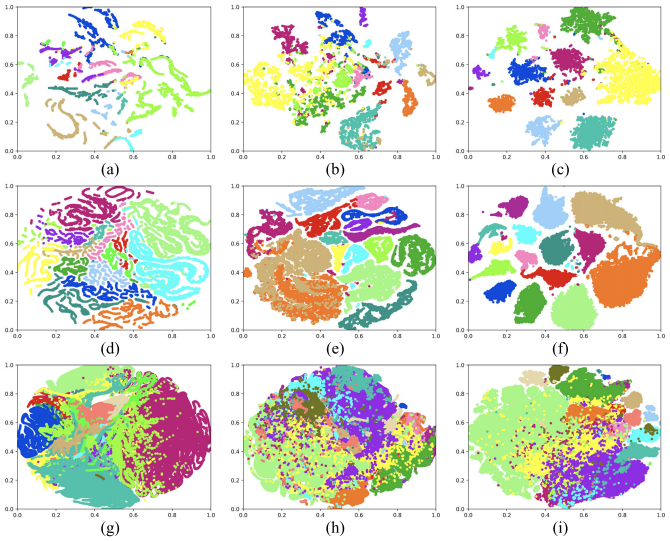


Fig. 9. Feature separability across various methods in three datasets. (a) WFCG. (b) CACFTNet and (c) GraphMamba for IP. (d) WFCG. (e) CACFTNet and (f) GraphMamba for SA. (g) WFCG. (h) CACFTNet. (i) GraphMamba for UH2018.

three datasets to validate the classification performance of GraphMamba. From Fig. 9, it can be observed that the features of GraphMamba are more aggregated, with larger distances between features of different classes. In contrast, WFCG and CACFTNet exhibit more overlapping and intersecting parts between features of different classes, with features of the same class being more scattered. This indicates that GraphMamba has a strong capability in spectral information mining, enabling the extraction of more discriminative information. This also suggests a broad application prospect for Mamba in HSI classification.

F. Ablation Experiment

To verify the effectiveness of the proposed HyperMamba and SpatialGCN, as well as the contributions of the GM, AutoRes Connection mechanism, and adaptive filtering mechanism to classification performance, we conducted ablation experiments on three datasets. We analyzed the impact of different components on the overall model by removing each component and comparing it with the baseline model. For simplicity, we represent the reduced methods as GraphMamba-v1 to GraphMamba-v5, as shown in Table V. The experimental setup is the same as in Section IV-C, using OA to report clustering accuracy, with the experimental results presented in Table V.

From the OAs across the three datasets, we can conclude: 1) first, the OA of GraphMamba-v1 and GraphMamba-v2 is significantly lower than that of GraphMamba-v3 to GraphMamba-v5, indicating that HyperMamba and Spatial-GCN play a crucial role in enhancing classification performance. Additionally, removing either of the two modules leads to a substantial decrease in classification accuracy, suggesting that Mamba and GCN effectively extract spatial–spectral information, and their combination results in greater performance improvement and 2) the experimental accuracy of GraphMamba-v3 to GraphMamba-v5 also shows varying degrees of decline, indicating that the three mechanisms—GM, AutoRes Connection, and Adaptive Filtering Mechanism—contribute significantly to improving classification accuracy, allowing for better joint extraction of spatial–spectral information for precise classification.

G. Hyperparameter Impact

In this section, we investigated the influence of different hyperparameters on the classification performance of the network. We employed a grid search strategy to find the optimal patch size and encoder depth settings. As shown in Fig. 10, we provided the OA for two different parameter values on three datasets.

As shown in Fig. 10, the overall classification performance is better when Patch = 11. Moreover, the classification results do not improve by blindly increasing the patch size, as too few land object details make it difficult to capture spatial features. In contrast, overly large patches are prone to mixing irrelevant land object information. Regarding the number of encoders, the overall performance is best when N = 6. Excessive network depth significantly consumes computational resources, making training more challenging. Therefore, choosing an appropriate number of encoders can extract spatial–spectral features effectively while meeting practical requirements.

H. Comparison of Runtime and Model Complexity

In this section, we compare the runtime and model complexity of all algorithms. Table VI presents the training and

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TABLE V

ABLATION STUDY RESULTS ON IP, SA, AND UH2018. THE OA INDEXES (%) ARE REPORTED, AND BOLD DENOTES THE BEST PERFORMANCES

Name		GraphMamba	GraphMamba-v1	GraphMamba-v2	GraphMamba-v3	GraphMamba-v4	GraphMamba-v5
Module	HyperMamba	✔	✘	✔	✔	✔	✔
	SpatialGCN	✔	✔	✘	✔	✔	✔
	Global Mask	✔	✔	✔	✘	✔	✔
	Automatic Residual Connection	✔	✔	✔	✔	✘	✔
	Adaptive Filtering	✔	✔	✔	✔	✔	✘
Dataset	IP	**95.99**	89.16	90.08	95.06	94.36	95.14
	SA	**97.23**	92.98	93.96	96.35	96.15	96.98
	UH2018	**86.45**	82.85	82.92	85.62	85.46	85.88

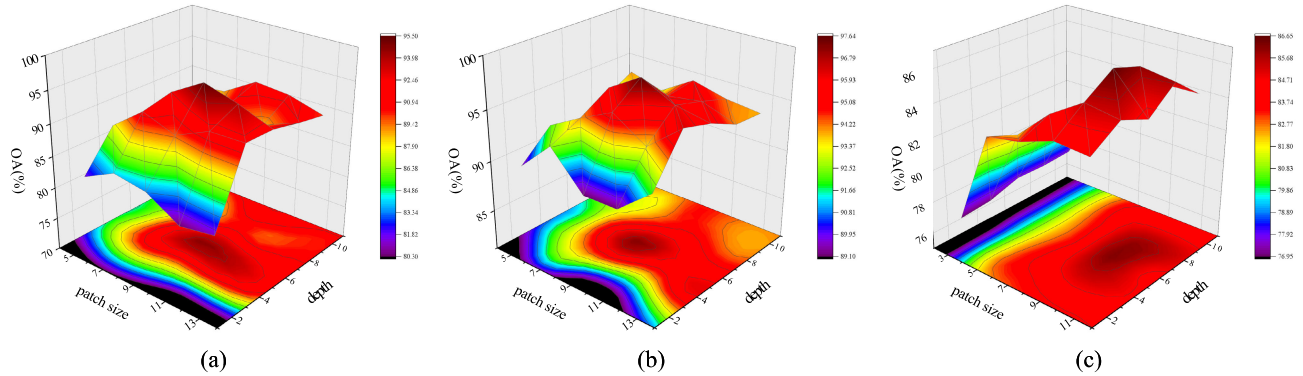


Fig. 10. Sensitivity of GraphMamba in different parameter settings (i.e., patch size and encoder depth) on (a) IP, (b) SA, and (c) UH2018.

TABLE VI

COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS. BOLD DENOTES THE BEST PERFORMANCES

Method	Running Time (s)						Complexity (FLOPs)
	IP		SA		UH2018		
	Train	Test	Train	Test	Train	Test	
3D-CNN	23.28	4.31	**10.81**	18.24	**43.21**	**35.53**	**0.69G**
DFFN	32.69	3.68	17.5	18.83	110.71	109.89	4.27G
RSSAN	21.85	**3.34**	13.33	11.86	72.49	89.25	0.85G
GAHT	38.07	4.69	23.38	21.6	96.02	100.87	4.72G
ABLSTM	58.22	4.66	37.38	35.81	99.28	115.8	5.32G
SF	385.14	73.26	188.16	316.45	1153.22	3107.71	1.50G
WFCG	**17.79**	3.54	38.28	**8.91**	49.3	39.15	5.98G
CACFTNet	57.42	3.74	28.14	9.75	456	137.85	6.55G
HSDBIN	214.39	5.73	157.25	12.46	264.78	67.74	2.61G
GraphMamba	102.03	5.65	97.32	19.48	127.94	52.98	**0.69G**

testing times of GraphMamba on three datasets, along with model complexity characterized by FLOPs. It can be observed runtime of GraphMamba is within an acceptable range. It is noteworthy that HyperMamba employs efficient parallel computing to reduce time, greatly enhancing computational efficiency. Additionally, GraphMamba exhibits lower complexity. Therefore, combining superior classification performance and lower training costs positions GraphMamba with broad prospects for industrial applications.

V. CONCLUSION

Spectral redundancy, complex spatial relationships, and the computational resource consumption of models have long been focal points in hyperspectral image classification. This article addresses these challenges by introducing an efficient graph structure learning vision Mamba for HSI classification, called GraphMamba. GraphMamba presents a universal paradigm for processing hyperspectral features, facilitating the extraction of discriminative features more conveniently and efficiently. Furthermore, we have designed the HyperMamba module to eliminate spectral redundancy, and the parallel computing approach significantly enhances computational efficiency and reduces overhead costs. SpatialGCN leverages adaptive context awareness and spatial filtering matrices to explore spatial features. Experimental results on three real datasets of varying scales demonstrate that, compared with various baseline methods, the proposed GraphMamba achieves more precise classification through the integrated use of spatial–spectral information, exhibiting superior performance.

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Aitao Yang received the B.S. degree in information engineering from Xi'an Institute of High Technology, Xi'an, China, in 2017. He is currently pursuing the Ph.D. degree in computer science and technology with Xi'an Hi-Tech Research Institute, Xi'an.

His main research interests include remote sensing information processing, hyperspectral image clustering, and computer vision.

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Min Li received the B.S. and M.S. degrees in computer science and technology from Xi'an Institute of High Technology, Xi'an, China, in 1991 and 1994, respectively, and the Ph.D. degree in communication and information from Xi'an Jiaotong University, Xi'an, in 1998.

She is currently a Professor with Xi'an Institute of High Technology. Her research interests include remote sensing, synthetic aperture radar (SAR) image processing, target detection and recognition, and feature extraction.



Yao Ding received the M.S. and Ph.D. degrees from the Key Laboratory of Optical Engineering, Xi'an Research Institute of High Technology, Xi'an, China, in 2013 and 2022, respectively.

He has published several journals in IEEE TRANSACTION ON GEOSCIENCE AND REMOTE SENSING (TGRS), *Information Sciences* (INS), *Expert Systems with Applications* (ESWA), *Defense Technology* (DT), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), and *Neurocomputing*. Furthermore, he has published three monographs, and six patents have been applied. His research interests include neural networks, computer vision, image processing, and hyperspectral image clustering.

Dr. Ding has received Excellent Doctoral Dissertations from the China Simulation Society and China Ordnance Industry Society in 2023. He also has received highly cited awards from Defense Technology (DT) Journal. At present, he has ten highly cited papers of ESI. He is a reviewer of TGRS, TNNLS, PR, JAG, and KBS. He served as a Guest Editor for *Forecasting*.



Leyuan Fang (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

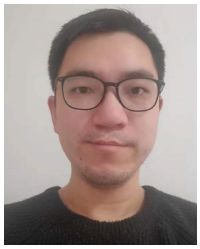
From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University, Durham, NC, USA. He is currently a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang was a recipient of one Second-Grade National Award at the Nature and Science Progress of China in 2019. He is an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Neurocomputing*.



Yaoming Cai (Member, IEEE) received the B.E. degree in information security and the Ph.D. degree in geoinformation engineering from China University of Geosciences, Wuhan, China, in 2016 and 2022, respectively.

He was a Visiting Scholar with Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany, from 2021 to 2022. He is currently an Associate Professor at the School of Information Engineering, Zhongnan University of Economics and Law, Wuhan. His research focuses on machine learning, multimodal learning, intelligent optimization, and remote image processing.



Yujie He received the B.S. degree in software engineering from Xidian University, Xi'an, China, in 2009, and the Ph.D. degree in intelligent information processing from Xi'an Research Institute of Hi-Tech, Xi'an, in 2016.

His research interests include infrared small target detection, sparse signal representation, and infrared target tracking.

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